Proceedings of the 7th International Conference on Trends in Electronics and Informatics (ICOEI 2023)
IEEE Xplore Part Number: CFP23J32-ART; ISBN: 979-8-3503-9728-4

Internet of Things (IoT) based Digital Helmet Design and Deployment

Mr. B Ravi Chandra
Department of Electronics and Communication Engineering
G. Pullaiah College of Engineering and Technology
Kurnool, India
ravichandraece@gpcet.ac.in

M.Pranaya Akshaya
Department of Electronics and Communication Engineering
G. Pullaiah College of Engineering and Technology
Kurnool, India
pranayaakshaya@gmail.com

B.Sravani
Department of Electronics and Communication Engineering
G. Pullaiah College of Engineering and Technology
Kurnool, India
sravanibadinehal143@gmail.com

Y.Sai Ashritha
Department of Electronics and Communication Engineering
G. Pullaiah College of Engineering and Technology
Kurnool, India
ashrithayakkaluru@gmail.com

B.K.Pavithra
Department of Electronics and Communication Engineering
G. Pullaiah College of Engineering and Technology
Kurnool, India
bkpavithra30@gmail.com

2023 7th International Conference on Trends in Electronics and Informatics (ICOEI) | 979-8-3503-9728-4/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICOEI56765.2023.10125847

***Abstract*—Each year around 1.3 million traffic accidents occur worldwide. Most of the accients occur due to two-wheelers. Driving while intoxicated, driving recklessly, and being fatigued from long distances are the main reasons for accidents. To avoid accidents and alert the rider, this study has designed and developed a smart helmet using Internet of Things (IoT) technology. The ignition switch is automatically set to the OFF position if a helmet isn't worn. Sensors are also used in designing a smart helmet. In this instance, a touch sensor is used to identify whether a motorcyclist is wearing a helmet or not, and an alcohol(MQ-3 gas)sensor is used to assess whether the rider is inebriated or not. The ignition locks on its own and sends a message to the certain mobile contacts if the rider is drunk or takes off the helmet while riding. The proposed smart application uses Blynk-2.0 cloud communication technology, where the mobile application automatically transmits the accident location to the police and the emergency contact information in the event of an accident. The accident vehicle's location can also be tracked by using the GPS device.**

Keywords— Cloud blynk 2.0, Global Positioning System (GPS), MQ-3 Sensors, IR Sensor, Liquid Crystal Display (LCD), DC 120V fan, Buzzer, Node MCU-ESP8266.

I. INTRODUCTION

Thousands of people are now seriously concerned about traffic accidents since so many young people are either killed or injured. According to the research, the majority of those killed in traffic accidents were between the ages of 15 and 34. A head injury is the main factor in the majority of fatal motorbike accidents.The rider should put on a smart helmet to preserve life and avoid a brain injury. Without the smart helmet, a vehicle's ignition will not turn on.The helmet is a requirement for motor vehicles under the 1988 Motor Vehicles Act (section 129).Every individual riding a two-wheeled vehicle must wear a helmet to secure their life, according to government rules.This technologically advanced helmet shields the wearer from the majority of negligence, including operating a vehicle without the necessary understanding of traffic regulations, maintaining a bike in good working order, and using a phone while driving, among other things.The failure to provide prompt and appropriate medical care as well as society's quick action in alerting the police and hospital are some of the primary causes of the increased fatality rate in accidents. In this scenario, numerous lives have been lost. Here, saving lives during prime time is important. Stop allowing time to claim anyone's life. The Smart helmet complies with government standards and aids in traffic enforcement.Riders on two-wheelers can utilize the equipment with total safety.[1]Motorcycles, also referred to as motorbikes or two-wheelers, are frequently utilized as alternative forms of transportation since they are less expensive than four-wheelers, but they are also the most dangerous.Driving too quickly or when intoxicated can also result in accidents.[20]Safety on the roads and security on bikes are people's top priorities in light of the astonishing IoT technology and rising urbanization.The failure to wear a helmet contributes to thousands of accidents every day, many of which result in fatalities or significant property damage.A helmet is one type of protective equipment used to shield the head from harm.The helmet specifically helps the skull protect the brain. Sometimes it's unable to locate the scene of the accident. A smart helmet can locate the scene of accidents,[18] saving lives and enhancing the safety of two-wheelers. The system is divided into three application categories: mobile, helmet, and automobile. When a mobile application notices an accident, it immediately contacts the cops and emergency contacts through the data with the accident site [2].Numerous safety regulations and guidelines have been established by the government for motorists to follow, yet many of them are rarely followed. The main goal of this endeavor is to create internet-connected smart helmets that increase two-wheeler riders' safety. The advanced MQ3 alcohol sensing,[25] IR sensor, and fall detection modules are included in the suggested smart helmet system. The MQ3 alcohol sensor is used in the circuit to prevent the bike from starting when the user has taken a significant amount of alcohol[15] If a motorcycle accident occurs, information is sent to the registered number using the GSM (Global System for Mobile Communication) and GPS (Global Positioning System) [23].Two basic safety elements are found in

Authorized licensed use limited to: VIT University. Downloaded on February 16,2026 at 16:59:39 UTC from IEEE Xplore. Restrictions apply.

motorcycle helmets. The exterior shell, which is commonly composed of plastic, fiberglass, or Kevlar, comes first, followed by the inner shell, which is normally constructed of polyphenylene foam. The rigid external shell's main function helps to stop the helmet or head from being penetrated and to stop the soft inner liner's structure from remaining stable during an impact [21]. Bikers using helmets suffered far fewer wirehead and neck accidents than motorcyclists without helmets[3].Every year, an alarming number of motorcycle and bike accidents happen on the roads of many nations. However, wearing a helmet can significantly lower the chance of accidents. For communication, there will be two primary components: the transmitter and the receiver. The bike that receives the RF will serve as the receiver, and the helmet will serve as the sender. Only after the rider fastens his or her helmet will the bike's engine turn on [4].A system for automatic accident detection using vision has been created [5]. Helmets are extremely important for preventing head injuries in event of an accident. The objective of this study is to create an electronic helmet that is smart, secure, and helmet essential for preventing accidents and saving lives. Information regarding the vehicle and the driver can be sent and tracked more easily with the help of this helmet.This research primarily focuses on warning drivers about approaching vehicle movements while changing lanes and it also helps lead drivers in the right direction [6]. According to the data on the world market for two-wheelers, the market is projected to grow to $125 billion by 2024. Approximately 1.35 million individuals every year pass away as a result of traffic accidents, according to WHO figures. And of those, more than half are motorcyclists, bikers, and pedestrians. The chance of fatal injuries can be reduced by 42%, and the risk of brain injuries can be reduced by 69% when the right helmets are used. The primary driver behind developing this intelligent helmet was safety. The goal of a prototype is to develop a smart technique for user safety that is integrated with sensors and modules [7]. The author draws information from the largest two-wheeler market database in the world. India has a high rate of accidents and head injuries in this. In India alone, motorbike accidents cause about 1 lakh fatalities each year, with a fatality rate ranging from 21.6% to 29.1% for every 100 accidents [8].25% of all traffic fatalities are related to alcohol, which is the key factor in many bike accidents[24]. The preventative measure's smart helmet component stops the relay from turning on the engine [9]. Applications for artificial intelligence are being created that make life simpler and are increasingly complex in how they learn and make judgments for humans [10]. The smart helmet is linked to the motorcycle engine. The goal of this article is to create a safety rider helmet prototype and forecast current conditions. Based on the IoT, a smart helmet is created that connects to the motorbike engine using a wireless module. Latitude and longitude coordinate link activities together via GSM and GPS [11]. This initiative is intended to save precious lives and is based on traffic in India. This system's primary goal is to direct people to the closest hospital and save lives [12].The author suggests using two different sets of role systems in the system. The motorcycle won't start if the helmet senses the alcohol in the atmosphere[17]. A 3-axis accelerator sensor is used to detect accidents. This design's goal is to make a system proposal. This aids in averting disasters of any type, and if such circumstances do arise, it identifies them and alerts the appropriate parties and the public so that the situation can be brought under control. Not only would texting nearby hospitals not be sufficient since it cannot prevent further accidents, but this method also satisfies this need [13]. To protect the head, a helmet is worn. Although there are several helmets for sale on the online and offline markets, they are not cost-effective and only focus on a limited number of factors, such as alcohol sensors. The goal of the model is to create a unique, innovative, and cost-effective multifunctional smart architecture for a helmet that will not only protect the rider but also often text their loved ones with updates on where they are and alert concerned parties if they are in an accident. It uses (Global Positioning System) GPS [22] GSM (Global System for Mobile Communication), IR, and alcohol sensors, among other things. Making the rider put on a smart helmet can help save his life in this way. The Internet of Things and other cutting-edge technologies will be utilized by this smart helmet (IoT). This work has been improved for users and is user-friendly, easy to use, and reasonably priced for everyone [14]. A good way to solve the issue is to utilize a smart helmet band. The smart helmet is a ground-breaking idea that increases the safety of motorcycling. (Global Positioning System) GPS[19] tracking allows us to determine the location. Developing nations like India are seeing a rapid rise in road accidents as a result of improved transportation technology and an increase in the number of automobiles on the road.

II. LITERATURE SURVEY

[1] A helmet with smart features using IoT Technology, IoT has made it possible for us to link daily gadgets together solely for the exchange of data. The wearing of a helmet while riding is currently required in several nations. In this article, I'll go through how the newest Internet of Things technology was used to make helmets smart. To make riding more convenient, this helmet has several features, including the ability to use navigation services, SOS signals in an emergency, and listen to music while traveling.

[2] Intelligent Helmets with Autonomous Headlamp Control the "Intelligent Safety Helmet for Motorcyclists" is designed to improve motorbike riders' level of road safety. In many countries, it is forbidden to ride without a helmet on. India serves as an example. The idea was inspired by the notion that motorcycle riders are growing more anxious as a result of an increase in fatal traffic accidents over time. This prototype will create automatic light technology for motorbike riders' safety. Here, focusing on headlamps that are capable of reacting to a rider's facial motion. It turns the headlights in the right direction by using a tiny electric motor built into the headlamp housing in conjunction with an accelerometer and other sensors.

[3] The Implementation and Analysis of Smart Helmets Accidents are a serious concern for everyone. As accidents occur frequently, efforts and measures are made to prevent and stop their effects of them. People in the environment frequently break the rules of the road because they don't give a damn about them. People also tend to challenge authority figures. To provide safety with luxurious and intelligent qualities, therefore, approach the problem from a different perspective. To make sure the cyclist is wearing the helmet, two modules-one on the helmet and one on the bike will cooperate. A radio frequency module enables wireless communication between the bike circuit and the helmet. The piezoelectric buzzer's capacity to detect speeding is improved by restricting the user's speed. Driving while

Authorized licensed use limited to: VIT University. Downloaded on February 16,2026 at 16:59:39 UTC from IEEE Xplore. Restrictions apply.

intoxicated can be prevented with the ALCHO-LOCK function. The ability of the accelerometer to identify accidents is increased by the use of a GSM module in the circuit, that is intended to automatically transmit a message to one individual and one concerned authority. A fog sensor is also used to increase vision when there is smog or fog. Using an additional feature known as HELMET, users can wirelessly have the required amount automatically taken out of their virtual wallet without having to physically stop and pay for it..

[4] Connect: An Accident Detection and notification Smart Helmet The Internet of Things(IoT)-based smart helmet's major objective is to provide people with a way to identify and report incidents. Sensors, Wi-Fi-enabled CPUs, and cloud computing infrastructures are used to construct the system. The CPU receives the accelerometer measurements from the accident detection system and continuously checks for irregular changes. When an accident occurred, the essential information is provided via a cloud-based service to the emergency contacts. The global positioning system is utilized to determine the location of the vehicle. The device suitably dubbed connects purports to relay accurate and timely information about accident detection and is developed by utilizing ubiquitous connectivity, which is a crucial component of smart cities.

[5] A highly secure smart helmet using the Internet of Things: This prototype's key aims are the detection, notification, and avoidance of accidents. The rider is supplied with exceptional levels of security and protection while also feeling comfortable with this helmet. Cloud-based services were connected over Bluetooth and Raspberry Pi 3. The image can be retrieved and transferred from the cloud through the helmet to the recipient's car and vice versa. Raspberry Pi will get orders from sensors. As a result, the recipient will get the order. A piece of software has been developed that uses Google Maps to locate the specific position. Cloud-based services will communicate with recipient contacts via messaging whose contact information is kept in databases. The majority of accidents are caused by reckless driving, driving while inebriated, using a cell phone while driving, and breaking traffic laws. Many people lose their lives as a result of accidents being reported too late, making it impossible to establish the specific GPS location of the accident scene. These are occasions when one is unable to notify of an accident that occurred. Not wearing a helmet is the main cause of many injuries, according to multiple victims.

[6]. This research provided a method for the rider's safe driving. To detect the rider's mental state and alcohol consumption, the smart helmet contains a breath alcohol sensitizer. Consider a circumstance where someone is employing a smart mp3 player to listen to music, where the music can be automatically changed for the safety of the rider.

[7]. Smart Helmet, Everyday items can now be connected to a network for the sole purpose of sharing data thanks to IoT technology. Today wearing a helmet while riding a vehicle is required in many nations. This study, explains how IoT technology was leveraged to make a helmet smart. This helmet includes several capabilities for the convenience of the rider, including the ability to use navigation services, listen to music while moving, and transmit SOS signals in an emergency.

III. RELATED WORK

The proposed strategies include preventing accidents by preventing the vehicle from starting if a person consumed alcohol. Everything works when the person is wearing a helmet. Some of the components that are being used to build a smart helmet are:

- IR sensor
- Alcohol sensor
- TTP223 - 1Touch sensor
- Buzzer
- DC 12V 2510 Fan
- GPS
- LCD
- Blynk 2.0

A. IR Sensor :

IR Sensor module has excellent ambient light adaption capabilities. It consists of a pair of infrared emitting tubes and a pair of infrared emitting tubes and a pair of infrared receiving tubes. The emitting tubes emit a specific frequency and encounter an obstacle-detecting direction (a reflecting surface), and the receiving tubes receive the reflected infrared. After processing by a comparator circuit, the output of the receiving tube isis a digital signal (a low-level signal)that is routed through a potentiometer knob to a LED. The sensor's detecting range may be adjusted using a potentiometer. It has minimal interference, is simple to build and operate, and is widely applicable for tracking black and white lines, avoiding obstacles, and other uses.

B. Features of IR Sensor:

- In addition to being driven directly to a 5V relay, connecting the sensor module's output port (OUT) to the microcontroller's IO port;
- Connections: GND-GND, OUT-IO, and VCC-VCC.
- The LM393 comparator is a stable device.
- The model's methods include the use of a module for a -5v Dc power source. When the electricity is turned on, the red power LED glows.
- With screw holes of 3mm, it is easy to install.
- Each module in the delivery has a threshold comparator voltage changeable through a potentiometer, in special cases; please do not adjust the potentiometer.

Authorized licensed use limited to: VIT University. Downloaded on February 16,2026 at 16:59:39 UTC from IEEE Xplore. Restrictions apply.

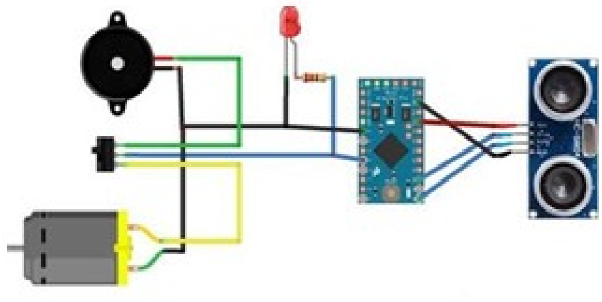

Fig. 1. IR sensor

C. *Alcohol Sensor:*

- This semiconductor gas sensor module, which is both inexpensive and simple to use, has both analog and digital output. The MQ3 Alcohol gas sensor is used in this module as an alcohol gas sensing element. It only needs the VCC and ground pins plugged in; no other components are needed.
- For digital output, an onboard potentiometer makes it simple to set the threshold value. You may quickly connect the MQ3 Alcohol sensor to any microcontroller, Arduino, or Raspberry Pi by using this module. This Gas sensor module can be used in a breathalyzer because it is alcohol sensitive. Benzine is similarly only weakly sensitive to the MQ3 sensor.

Specifications of the MQ3 GAS Sensor Module :

- The interface type is Analog and Digital.
- The power supply is 5 volts.
- Extremely High Sensitive to Alcohol and Very low Sensitive to Benzine.
- Low cost
- Long life and Onboard power indication.

Fig. 2. Alcohol sensor

D. *TTP223 - 1 Channel Capacitive Touch Sensor*

- It is simple to incorporate capacitive thanks to the TTP223-1 Channel Capacitive Touch Sensor Module, which employs the touch sensing IC TTP223(Time Triggered Protocol) to sense the touch input. The purpose of the touching-detecting IC is to replace conventional direct button Keys with various pad sizes. There is only one touchpad that is fully operational to sense input when the module is powered by two 5.5V DC batteries.
- The interface for a single-channel capacitive touch sensor module is quick and simple. It can be used without, in addition to, or in place of an Arduino or microcontroller. The sensor detects the change in capacitance when a capacitive load, such as a human hand, approaches the sense pad and riggers the switch.
- When a finger is placed on the right spot, the module output is high and the power consumption is low. If the spot is not touched for 12 seconds, the module switches to low-power mode.
- The module can be put on surfaces made of non-metallic materials, glass, or plastic.

Fig. 3. Touch sensor

E. *Buzzer:*

- A buzzer or beeper is an auditory signaling device that can be electromechanical, piezoelectric, or mechanical (piezo for short). Alarm clocks, timers, and confirmation of human input like a mouse click or keyboard are common applications for buzzers and beepers.
- We are making use of a piezoelectric buzzer. Based on the reverse of the fundamental idea is to apply an electric voltage across a piezoelectric material to generate pressure fluctuation or stain.
- Regardless of the voltage variation provided to it, the buzzer makes the same loud sound.
- Here we use the 5V buzzer.

Fig. 4. Buzzer

F. *DC 12V 2512 Brushless Fan Motor:*

This little fan is an exhaust cooling fan. As little as a palm, this fan is. It operates at 12V and 0.13A. It can operate without issue with a standard 12V battery. This little fan has a speed range of 6800 to 13000 rpm (revolution per minute). The resin and plastic used to construct the fan's body are mixed. The combination gives the fan stability and insulation. Because of how it was made, it is both lightweight and sturdy enough to withstand some falls to the

Authorized licensed use limited to: VIT University. Downloaded on February 16,2026 at 16:59:39 UTC from IEEE Xplore. Restrictions apply.

ground. Therefore, you have come to the proper place if you're looking for a single fan that perfectly balances strength and insulation.

1) Features:

- Very little weight.
- Compact enough to fit in the hand.
- Runs on 12V DC.
- Can tolerate high room temperatures.

2) Specifications:

- Working voltage: 12V
- Model number: 2510
- Lightweight
- Durable.

Fig. 5. . DC 12V Fan

G. GPS (Global Positioning System):

GPS stands for the Global Positioning System which consists of the following three components and allows anyone to constantly acquire position information anywhere in the world.

- GPS satellites` space segment
- Several GPS satellites are in operation on six orbits around the planet at a height of roughly 20,000 km, traveling around the planet every 112 hours.
- Control segment (Ground control station):
- Ground control stations make up the control segment.
- Ground control stations are responsible for monitoring, managing, and maintaining satellite orbit to ensure that GPS timing and satellite orbital deviation are both within acceptable limits.
- User segment (GPS receiver):
- The user group (GPS receivers)

H. Control segment (Ground control stations):

Satellite orbit is monitored, controlled, and maintained by ground control stations to ensure that GPS timing and satellite orbital variation are both within acceptable limits. User segment (GPS receivers)

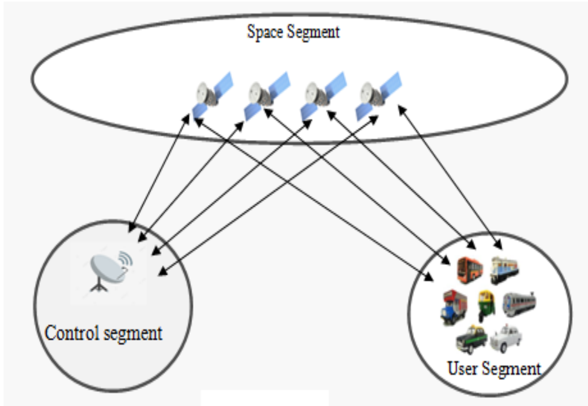

Fig.6. GPS

I. LCD (Liquid Crystal Display):

The parallel interface of LCDs necessitates simultaneous manipulation of many interface pins by the microcontroller to control the display. Hitachi-compatible LCDs support both 4-bit and 8-bit control modes. For the 4-bit mode and the 8-bit everything mode, respectively, the Arduino requires seven and eleven I/O pins. Almost for text displays on the screen can be done in 4-bit mode. The example below demonstrates how to operate a 2x16 LCD in 4-bit mode.

Fig. 6. Liquid Crystal Display

J. Microcontroller:

The microcontroller Node MCU-ESP8266 has Wi-Fi capabilities. This IoT platform is a free source. Using Hayes-style commands, this tiny device enables microcontrollers to join a

Wi-Fi network and create straightforward TCP/IP connections. Node MCU alludes to the firmware that is set to default. The scripting language employed by this firmware is Lua. It uses the ESP8266 processor and XTOS as its operating system. It contains 4MB of storage and 128KB of RAM.USB is used to supply the controller with power.

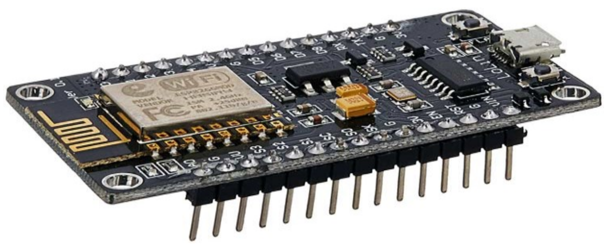

Fig. 7. Node MCU

Authorized licensed use limited to: VIT University. Downloaded on February 16,2026 at 16:59:39 UTC from IEEE Xplore. Restrictions apply.

K. *CLOUD-Blynk 2.0:*

All of this data is transmitted over the cloud to family members, emergency services, etc.

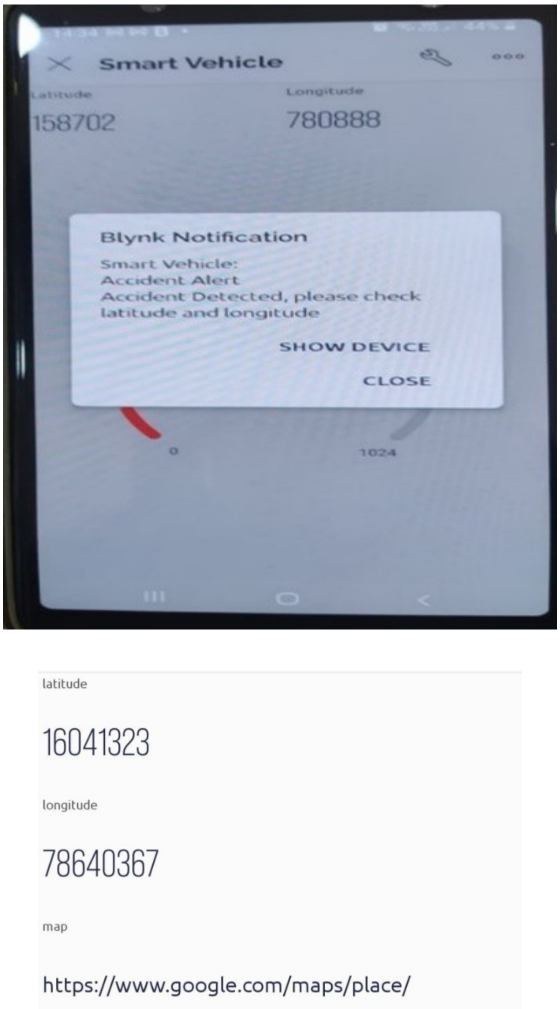

Fig. 8. Blynk 2.0 Cloud

L. *Limitations and issues:*

- The price of the helmet rises due to the use of Arduino in the system.
- Instead of using Arduino, Microcontroller is replaced as it consumes less power.

IV. PROPOSED SYSTEM:

The suggested technique seeks to lessen accident fatalities while also preventing motorbike accidents. Accidents are reported along with their location to family members, police enforcement, and the closest hospital. High-accuracy accident detection algorithms automatically record, identify, and report accidents right away. Cyclists can drive more safely thanks to the driving data. This technique instills the practice of helmet use in motorcycle riders. A motorbike trip would be safer and more protected with a smart helmet.

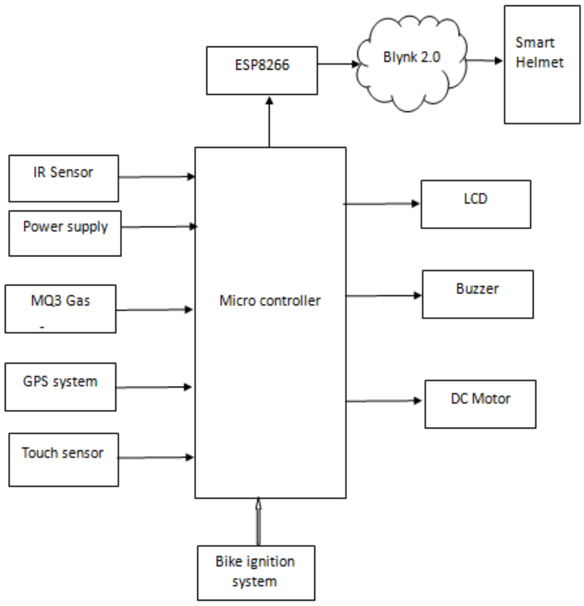

Fig. 9. Block diagram

Here, we develop a system that constantly checks two conditions before the rider turns on the bike's ignition. A touch sensor determines the first criterion, which is whether the rider is wearing a helmet or not. The second requirement is the use of an alcohol sensor to identify the presence of alcohol in the rider's breath.

Data is gathered and processed for the biking part by the helmet's microcontroller unit. Current latitude and longitude are retrieved by the connected GPS receiver, which changes them to the predefined value with the predefined periodicity when the bike is started, and then starts the engine. In the event of an accident, an IR sensor detects the collision, and an alarm message is sent via IoT mode to a preset number as well as to the closest police station and ambulance.

Authorized licensed use limited to: VIT University. Downloaded on February 16,2026 at 16:59:39 UTC from IEEE Xplore. Restrictions apply.

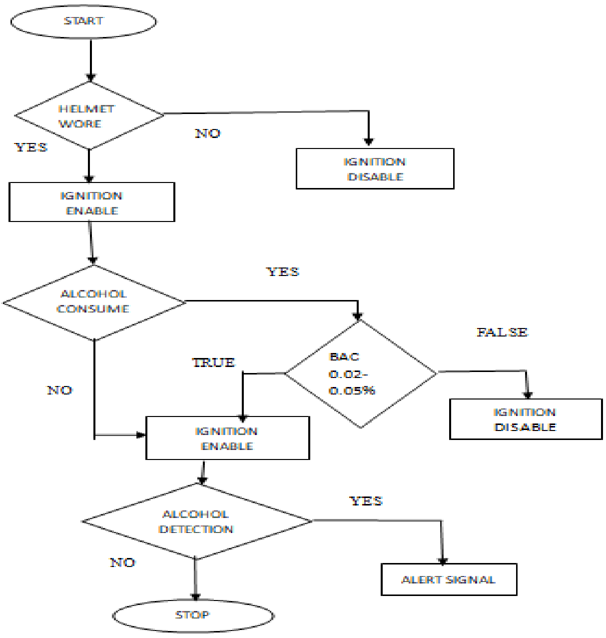

Fig. 10. Flow Chart

The smart helmet prototype is shown in this publication. A network of integrated sensors is used in the design of a smart helmet for accident or alcohol detection and reporting. The helmet is designed to do two different tasks.

1. Motorcycle ignition and alcohol detection

2. The generation of instant alert signals and accident detection.

Similar to a seat belt reminder in a car, the helmet is made to detect if the rider is wearing one or not. If the rider is not, the helmet will sound an alert to remind them to do so (it will only alert when the bike ignition is on). If the rider's breath contains more alcohol than is permitted, the helmet is designed to detect it. This will turn off the car's ignition. The MQ-3 alcohol sensor, which is installed on the helmet, measures the rider's breath alcohol concentration. If the BAC (Blood Alcohol Content) is greater than 0.02% to 0.05% in 100ml of blood, the helmet will start the engine and it will stop the ignition of the car. Additionally, the helmet is built to detect accidents and immediately notify emergency contacts. At regular intervals, the alert signals are delivered again and again.

V. ADVANTAGES

- Accidents in remote locations can be quickly identified, and medical assistance can be given.
- Using an alcohol detector will lower the likelihood of accidents by simply discouraging drunk driving.
- Affordable and simple to swap out with other sensors.
- Smart helmets are in charge of detecting drowsiness, speed, blinking for accident avoidance, and they are responsible for sending out warning notifications to accidents.

VI. APPLICATIONS

- It applies to real-time safety systems.
- The entire circuit can be built into a tiny VLSI chip and integrated into the helmet and bike unit. We can fit the entire circuit into a tiny VLSI chip and integrate it into the bike and helmet system.
- It might be designed for a safety system with low electrical requirements.
- By substituting a seat belt for a helmet, this safety mechanism could be enhanced even further when used in a car or other vehicle.
- Smart helmet includes information about their functions,sensors,microcontrollers,wireless communication technology, and other details.

VII. RESULT AND EXPECTED IMAGE

- The helmet ensures the motorcycle riders' safety and also takes the required actions to lower accident causes. The technology won't let the rider start the motorbike if any safety procedures are broken. The distance between the mouth and the sensor affects the accuracy of the alcohol sensor. When the distance is greater than 15cm, the sensor's value of 150mg/L shows the rider has a low alcohol level. The rider's blood alcohol content increases as the distance decrease to less than 6 cm when the alcohol level hits>350mg/L. This technique uses a smartphone with a GPS built-in. Poor mobile reception, fast mobility, and old GPS modules are the main causes of decreasing GPS accuracy. On the other hand, a sensor detects accidents.

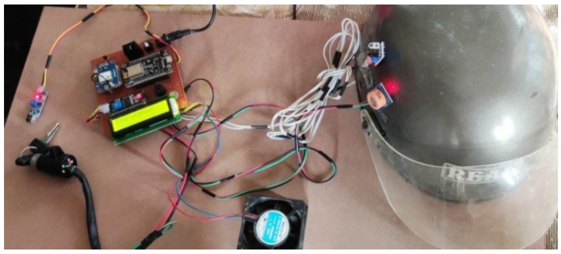

Fig 11(a)**: An experimental smart helmet configuration**

Fig 11(b): **An experimental smart helmet configuration**

VIII. CONCLUSION

In the proposed model, the ignition switch is automatically set to the off position if the helmet is not worn. Touch Sensors, IR sensors, and MQ-3 (alcohol) Sensors are employed in helmets. Touch Sensor is used to determine whether or not the rider is wearing a helmet. Blynk 2.0 cloud communication technology is employed for smart application purposes, and it broadcasts the accident's location to the

Authorized licensed use limited to: VIT University. Downloaded on February 16,2026 at 16:59:39 UTC from IEEE Xplore. Restrictions apply.

police and the emergency contact that was saved in contacts. The longitude and latitude that are attached to the bike are used to pinpoint the exact location of the accident vehicle. A medical emergency can be rescued by an ambulance by learning the location.

IX. FUTURE SCOPE

- Various bioelectric sensors installed in the helmet enables to track the rider's statistics and a range of activities.
- In the future, we will be able to use the invisible camera to record all accident-related data, which will aid in subsequent investigations.
- Solar power can be used to charge electric vehicles and mobile gadgets while riding two wheels.
- A future self-driving motorcycle could be built using artificial intelligence, protecting the rider and averting collisions.

REFERENCES

[1] Divyasudha N., Arulmozhivarman. P. and Rajkumar. E.R, "Analysis of smart helmets and designing an IoT based smart helmet: A cost-effective solution for riders," 2019 1st International Conference on Innovations in Information and Communication Technology (ICIICT), Chennai, India,2019,pp. 1-4 https://doi.org/10.1109/ICIICT1.2019.8741415.

[2] Md.Atiqur. Rahman, S.M. Haussamen, Ishman. Rahman, Toufiq. Ahmed and Abid. Ahsan, "IoT Based Smart Helmet ad Accident Identification System," 2020 IEEE Region 10 Symposium(TENSYMP), Dhaka, Bangladesh, 2020, pp.14-17, http://doi.org/10.1109/TENSYMP50017.2020.9230823.

[3] P.Koteswara Rao and Sai, P Tarun and Kumar, N Vinay and Sagar, SK Yusuf Vidya, Design And Implementation Of Smart Helmet Using IoT(January 19, 2020). International Conference of Advance Research & Innovation (ICARI) 2020 http://dx.doi.org/10.2139/ssrn.3643615.

[4] Mohammed Ehsanul Alim, Sarosh. Ahmad, Marzieh Nagahdi. Dorabati and Ihab Hassoun, "Design & Implementation of IoT Based Smart Helmet for Road Accident Detection," 2020 11th IEEE Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON), Vancouver, BC, Canada, 2020, https://doi.org/10.1109/IEMCON51383.2020.9284820.

[5] Harshit Chauhan, Ramsha Rizwan and Mehtab. Fatima," IoT Based Fall Detection of a Smart Helmet," 2022 7th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India,2022,pp.407-412, https://doi.org/10.1109/ICCES54183.2022.99835890.

[6] Dubey, Sourav; Meghana, K. ;likitha, Mamidala; Gupta; Kajal; Balaji, R.(2020). An experimental study on advanced lane changing signal assist technology with a smart helmet. https://doi.org/10.1016/j.matpr.2020.08.361.

[7] Ninad V. Joshi, SumedhaP.Joshi, Malhar S. Jojar, Neel S. Joshi and Anjali R. Askhedkar, "Design and Finite Element Analysis of IoT Based Smart Helmet," 2020 IEEE International IoT, Electronics and Mechatronics Conference (IEMTRONICS), Vancouver, BC, Canada,2020, https://doi.org/10,1109/IEMTRONICS51293.2020.9216393.

[8] M Lokeshwaran, S. P. Nikhit Mathew and A. Joshuva, "Raphael-The Smart Helmet,"2020 International Conference on Wireless Communication Signal Processing and Networking (WiSPNET), Chennai, India, 2020, https://doi.org/10.1109/WiSPNET48689.2020.9198463.

[9] Gurupreet. Singh Chhabra, Monika Verma, Khushi. Gupta, Abhinandan. Kondekar, Siddarth Choubey and Abha Choubey, "Smart Helmet using IoT for Alcohol Detection and Location Detection System." 2022 4th International Conference on Inventive Research in Computing Applications (ICIRCA), Coimbatore, India, 2022, https://doi.org/10.1109/ICIRCA54612.2022.9985543.

[10] Archana D., Boomija G., Manisha J., and Kalaiselvi V.K.G., "Mission on! Innovations in bile systems to provide a safe ride based on IoT," 2017 2nd International Conference on Computing and Communications Technologies (ICCCT),Chennai, India, 2017, https://doi.org/10.1109/ICCCT2.2017.7972296.

[11] Somantri and I.Yustiana, "Smart Helmet Integrated with Motorcycles to Support Rider Awareness and Safety Based Internet of Things,"2022 International Conference on ICT for Smart Society (ICISS), Bandung, Indonesia, 2022, https://doi.org/10.1109/ICISS55894.2022.9915262.

[12] A. Jesudoss, R.Vybhavi, and B. Anusha, "Design of Smart Helmet for Accident Avoidance,"2019 International Conference on Communication and Signal Processing (ICCSP), Chennai, India, 2019,pp.0774-0778 https://doi.org/10.1109/ICCSP.2019.8698000.

[13] S. Nanda, H. Joshi, and S. Khairnar, "An IoT Based Smart System for Accidents Prevention and Detection," 2018 Fourth International Conference on Computing Communication Control and Automation (ICCUBEA), Pune, India, 2018, https://doi.org/10.1109//ICCUBEA.2018.8697663.

[14] Selvakumar.R, B. Sai Raghavendra, B. Murali Krishna, K. Thanush Reddy,K. Sasi Kumar, "Helmet Rider Safety in a Cost- Effective Way with a Smart Helmet", 2021 3rd International Conference on Advance in Computing, Communication Control and Networking (ICAC3N), Greater Noida, India, 2021, https://doi.org/10,1109/ICAC3N53548.2021.9725567.

[15] S. Johnpaul, C. Thirumalai Selvan, and P.J. Raguraman, "IoT Based Smart Helmet System for Accident Prevention," 2022 International Conference on Edge Computing and Application (ICECAA), Tamil Nadu, India, 2022, https://doi.org/10.1109/ICECAA55415.2022.9936313.

[16] Bindu Sebastian Priyanka k.p, HridhyaKuttikrishan, Smart Helmet International Journal of Technology & Advanced Engineering, Volume 5, Issue: 12, December 2015.

[17] Harish Chandra Mohanta, Rajat Kumar Mahapatra and JyotirmayeeMuduli, "Anti-Theft Mechanism Framework with Accidental Avoidance and Cabin Safety Framework for Automobiles", International Referred Journal of Engineering and Science (IRJES), vol.3, no.4,pp.56-62,June2014.

[18] Professor Chitte p.p., SalunkeAkshay S., Thorat Aniruddha, N Bhosale, Smart Helmet & Intelligent Bike System, International Research Journal of Engineering and Technology (IRJET) Volume: 03 Issue:05, May 2016.

[19] Mr. P. Dileep Kumar1, Dr . G. N Kodanda Ramaiah2, Mr. A. Subramanyam3, Mrs. M Dharni, "A Solar Powered Smart Helmet With Multi features" in International Diary of Engineering Inventions vol.4,pp:06-11,June 2015

[20] Manjesh N, Sudharshan Raj, Smart Helmet using GSM&GPS Technology for Accident Detection and Reporting System, International Journal of Electrical and Electronics Research,2,4(2014).

[21] Prudhvi Raj R, Sri Krishna Kanth, Bhargav Aditya Bharath K,(2014)" Smart -tec Helmet" Electrical and Electronics Engineering, GITAM University, Rushikonda, Visakhapatnam, India. Advance in Electronics and Electric Engineering 4: 493-498.

[22] Sreenithy Chandran, Sneha Chandrasekar, N Edna Elizabeth " Konnet: An Internet of Things (IoT) based smart helmet for accident detection and notification" 2016 IEEE Annual India Conference(INDICON).

[23] Rashmi Vashisth, Sanchit Gupta, Aditya Jain, Sarthak Gupta, Sahil, Prashant Rana "Implementation and analysis of smart helmet" 2017 4th International Conference on Signal Processing, Computing, and Control (ISPCC) https://doi.org/10.1109/ISPCC.2017.8269660)

[24] D.Selvathi, P. Pavithra, T. Preethi "Intelligent Transportation System for Accident Prevention and Detection" 2017 International Conference on Intelligent Computing and Control Systems(ICICCS) https://doi.org/10.1109/ICCONS.2017.8250761

[25] SayanTapadar, Shinjini Ray; Himadri, NathiSaha; Arnab, Kumar Saha, Robin Karlose "Accident and alcohol detection in Bluetooth enabled smart helmets for motorbikes" 2018 IEEE 8th Annual Computing and Communication Workshop and Conference(CCWC). https://doi.org/10.1109/CCWC.2018.8301639

Authorized licensed use limited to: VIT University. Downloaded on February 16,2026 at 16:59:39 UTC from IEEE Xplore. Restrictions apply.